

DS	Session	Tree 2	Question Information	Level 1	Challenge 71
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Question description

Given two rooted trees, your task is to find out if they are *isomorphic*, i.e., it is possible to draw them so that they look the same.

Constraints

- $1 \leq t \leq 1000$
- $2 \leq n \leq 10^5$
- the sum of all values of n is at most 10^5

Input

The first input line has an integer t : the number of tests. Then, there are t tests described as follows:
The first line has an integer n : the number of nodes in both trees. The nodes are numbered $1, 2, \dots, n$, and node 1 is the root.
Then, there are $n-1$ lines describing the edges of the first tree, and finally $n-1$ lines describing the edges of the second tree.

Output

For each test, print "YES", if the trees are isomorphic, and "NO" otherwise.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
2
3
1 2
2 3
1 2
1 3
3
1 2
2 3
1 3
3 2
```

Test Case 2

INPUT (STDIN)

```
2
3
1 2
2 3
1 3
3 3
3
1 2
2 3
1 3
3 3
```

EXPECTED OUTPUT

NO
YES

EXPECTED OUTPUT

NO
NO

▼ Mandatory Test Cases

Test Case 1

KEYWORD

```
while(t--)
```

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

480

Test Case 3

NLOC

65